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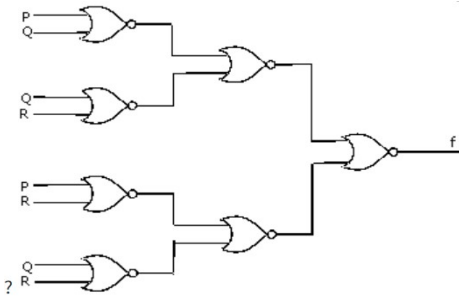
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RV COLLEGE OF ENGINEERING
(Autonomous Institution affiliated to VTU)
II Semester B.E. June/July -2026 Examinations
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
PRINCIPLES OF ELECTRONICS ENGINEERING
(2025 SCHEME)

*Time: 03 Hours**Maximum Marks: 100**Instructions to candidates:*

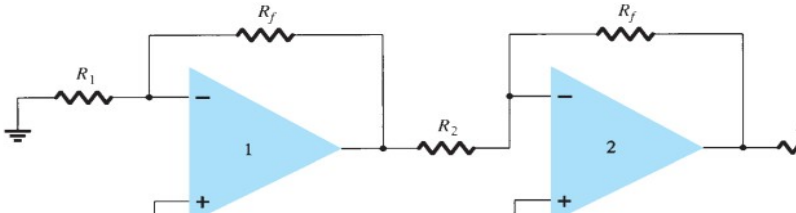
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 and 3 is compulsory. Answer any one full question from 4 or 5, 6 or 7, and 8 or 9.

PART-A

1	1.1	Determine the current through a PN junction diode at 37° C, If the forward voltage drop across the diode is 650mV and its reverse saturation current is 8nA. Assume $\eta=2$.	2
	1.2	A diode has a forward current of 0.5mA and its forward voltage is 500mV. Determine its reverse saturation current. Assume $\eta=2$.	2
	1.3	Photodiodes are operated in _____ bias to detect light efficiently.	1
	1.4	How many cells are present in a 3-variable K-map?	1
	1.5	If the differential gain of an amplifier is $A_d = 100,000$ and the CMRR is 80dB, find the common-mode gain A_{cm} .	1
	1.6	A non-inverting amplifier uses resistors $R_1 = 1k\Omega$ and $R_f = 9k\Omega$. Calculate the voltage gain (A_v) of the amplifier.	1
	1.7	What is the Boolean expression for the output f of the combinational logic circuit of NOR gates shown in figure 1.7?  Figure 1.7	1
	1.8	How many AND, OR and EXOR gates are required for the configuration of full adder?	1

PART-B

UNIT-I			
2	a	Write the Shockley's diode current equation and plot the diode characteristics. Explain each parameter of equation and V-I curve.	06
	b	Explain the working principles and applications of Photo diodes and LEDs.	06
	c	Draw the block diagram of a DC power supply and explain the function of each block.	06

UNIT-II			
3	a	With neat diagrams, explain the operation of the RC Phase Shift Oscillator. Derive the expression for frequency of oscillation and explain how the phase shift of 180° is achieved.	06
	b	State and explain the Barkhausen Criterion for sustained oscillations. Why is it necessary for oscillator design?	04
	c	Calculate the output voltage using the circuit of Fig. 3c for resistor components of value $R_f = 470 \text{ k}\Omega$, $R_1 = 4.3 \text{ k}\Omega$, $R_2 = 33 \text{ k}\Omega$, and $R_3 = 33 \text{ k}\Omega$ for an input of $80 \mu\text{V}$.  Fig. 3c	08

UNIT-III			
4	a	Using Boolean algebra theorems and properties, simplify the following Boolean expression step by step: $F(A, B, C, D) = \overline{A}BC + A\overline{B}D +$ Show each simplification step clearly and obtain the minimal SOP expression. Also draw the logic circuit using basic gates (AND, OR, NOT) for the simplified result.	06
	b	Design a Half adder and Full Adder circuit and Elaborate its working and give its (i) Circuit Diagram, (ii) Truth Table	08
	c	Explain the Canonical and Standard Forms of Boolean functions with suitable examples. Clearly differentiate between Canonical Sum of Minterms (SOP) and Canonical Product of Maxterms (POS)	04
OR			
5	a	Design a Full adder using two half adders and also derive the expression for sum and carry, along with its circuit diagram.	08
	b	Solve the following Expression Using K Map $F(A, B, C, D) = \sum m(0, 1, 2, 5, 7, 8, 9, 10, 13, 15)$	06

	c	State and explain DeMorgan's Laws using Boolean expressions.	04
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UNIT-IV			
6	a	Define amplitude modulation. Describe its process and write the mathematical expression for its waveform. Derive the expression for the total transmitted power of an AM signal in terms of carrier power and modulation index.	06
	b	Differentiate between Micro Processor and Micro controller	06
	c	Elaborate and explain the Von Neumann and Harvard architectures with suitable block diagram.	06
OR			
7	a	With a neat block diagram of Digital Communication system explain the working of each block.	06
	b	An audio frequency signal $10\sin 2\pi \times 500t$ is used to amplitude modulate a carrier of $50\sin 2\pi \times 10^5t$. Calculate (i) Modulation Index , (ii) Side Band Frequencies (iii) Amplitude of Each Side band frequencies , (iv) Bandwidth required (v) Total Power Delivered	06
	c	Differentiate between the RISC and CISC Architecture	06
UNIT-V			
8	a	Explain the key characteristics of embedded systems and classify them based on performance and complexity. Discuss one example from each class with its typical applications.	06
	b	Describe the major design metrics (Power, Size, Cost, Performance) and constraints in embedded systems development. Illustrate with a real-world trade-off scenario, such as in battery-powered devices.	06
	c	Outline the basics of Embedded Linux, including its kernel features, bootloader process, and filesystem support. Compare its advantages over bare-metal programming for IoT applications.	06
OR			
9	a	Define Edge Computing, explaining its need, typical architecture (including fog and cloud layers), and provide two practical examples. Highlight how it addresses latency issues in traditional cloud computing.	08
	b	Describe the steps to build an IoT system using ESP32, integrating cloud platforms like ThingSpeak or Blynk for data visualization. Illustrate with a case study on edge-based predictive maintenance or wearable health monitors.	10

Signature of Scrutinizer:

Signature of Chairman

Name:

Name:

Course Code:EC113ATC.

Course Title: **PRINCIPLES OF ELECTRONICS
ENGINEERING****PART-A**

Q.No	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10
B T	3	3	1	2	3	3	3	2		
COs	2	2	1	1	2	3	4	3		

PART-B

Question No		B T Levels	COs addressed		Question No		BT Levels	COs addressed
2	a	2	1		3	a	2	1
	b	2	1			b	1	2
	c	2	1			c	3	3
4	a	4	3		5	a	4	3
	b	3	4			b	3	4
	c	2	3			c	2	2
6	a	2	3		7	a	1	3
	b	2	2			b	3	2
	c	1	2			c	2	1
8	a	1	4		9	a	1	4
	b	2	1			b	2	4
	c	1	4			c		
10	a				11	a		
	b					b		

Signature of Scrutinizer:

Signature of Chairman

Name:

Name:

